

$${}^0T_2 = {}^0T_1 {}^1T_2 = \begin{pmatrix} c_1 c_2 & -s_1 c_2 & -s_1 & 0 \\ s_1 c_2 & -c_1 c_2 & c_1 & 0 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^3T_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & L \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0T_3 = {}^0T_2 {}^2T_3 = \begin{pmatrix} c_1 c_2 c_3 - s_1 s_2 c_3 & -c_1 c_2 s_3 - s_1 c_2 c_3 & -s_1 & c_1 c_2 a_2 - s_1 d_3 \\ s_1 c_2 c_3 - c_1 s_2 c_3 & -s_1 c_2 s_3 - c_1 c_2 c_3 & c_1 & s_1 c_2 a_2 + c_1 d_3 \\ -s_2 c_3 - c_2 s_3 & s_2 s_3 - c_2 c_3 & 0 & -s_2 a_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0T_4 = {}^0T_3 {}^3T_4 = \begin{pmatrix} c_1 c_2 c_3 - s_1 s_2 c_3 & -c_1 c_2 s_3 - s_1 c_2 c_3 & -s_1 & (-c_1 c_2 s_3 - s_1 c_2 c_3)L + c_1 c_2 a_2 - s_1 d_3 \\ s_1 c_2 c_3 - c_1 s_2 c_3 & -s_1 c_2 s_3 - c_1 c_2 c_3 & c_1 & (-s_1 c_2 s_3 - c_1 c_2 c_3)L + s_1 c_2 a_2 + c_1 d_3 \\ -s_2 c_3 - c_2 s_3 & s_2 s_3 - c_2 c_3 & 0 & (s_2 s_3 - c_2 c_3)L - s_2 a_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$J(\vec{\theta}) = \begin{pmatrix} \frac{\partial p_x(\vec{\theta})}{\partial \theta_1} & \frac{\partial p_x(\vec{\theta})}{\partial \theta_2} & \frac{\partial p_x(\vec{\theta})}{\partial \theta_3} \\ \frac{\partial p_y(\vec{\theta})}{\partial \theta_1} & \frac{\partial p_y(\vec{\theta})}{\partial \theta_2} & \frac{\partial p_y(\vec{\theta})}{\partial \theta_3} \\ \frac{\partial p_z(\vec{\theta})}{\partial \theta_1} & \frac{\partial p_z(\vec{\theta})}{\partial \theta_2} & \frac{\partial p_z(\vec{\theta})}{\partial \theta_3} \\ \xi_1^0 \vec{z}_1 & \xi_2^0 \vec{z}_2 & \xi_3^0 \vec{z}_3 \end{pmatrix}$$

$$= \begin{pmatrix} (s_1 c_2 s_3 + s_1 s_2 c_3)L - s_1 c_2 a_2 - c_1 d_3 & (c_1 s_2 s_3 - c_1 c_2 c_3)L - c_1 s_2 a_2 & (-c_1 c_2 c_3 + c_1 s_2 s_3)L \\ (-c_1 c_2 s_3 - c_1 s_2 c_3)L + c_1 c_2 a_2 - s_1 d_3 & (s_1 s_2 s_3 - s_1 c_2 c_3)L - s_1 s_2 a_2 & (-s_1 c_2 c_3 + s_1 s_2 s_3)L \\ 0 & (c_2 s_3 + s_2 c_3)L - c_2 a_2 & (s_2 c_3 + c_2 s_3)L \\ 0 & -s_1 & -s_1 \\ 0 & c_1 & c_1 \\ 1 & 0 & 0 \end{pmatrix}$$